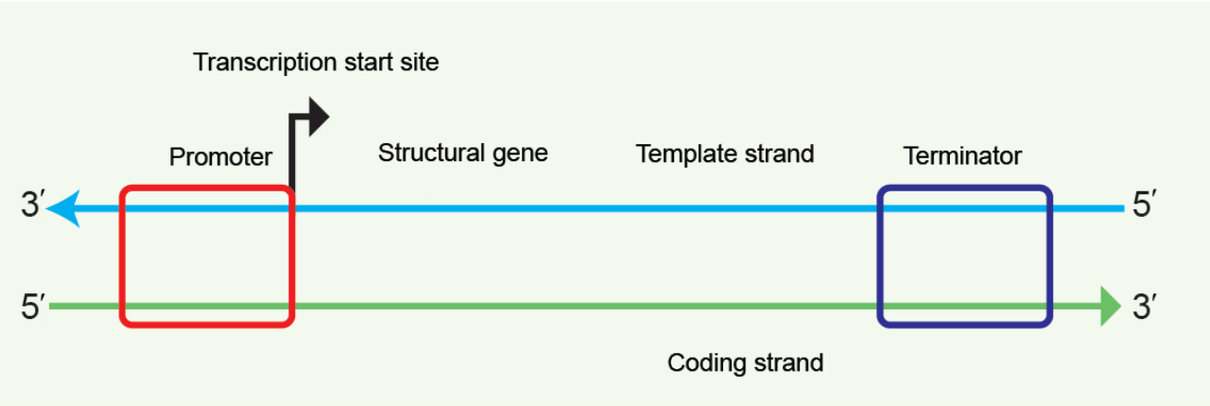


Fig. 5. 7 Schematic structure of a transcription unit

prokaryotes this region is called **Pribnow box**. Besides promoter, eukaryotes also require an enhancer.

The two strands of the DNA in the structural gene of a transcription unit have opposite polarity. DNA dependent RNA polymerase catalyses the polymerization in only one direction, the strand that has the polarity 3'⟶5' acts as a template, and is called the **template strand**. The other strand which has the polarity 5'⟶3' has a sequence same as RNA (except thymine instead of uracil) and is displaced during transcription. This strand is called **coding strand** (**Fig. 5.7**).

The structural gene may be **monocistronic** (eukaryotes) or **polycistronic** (prokaryotes). In eukaryotes, each gene transcribes a single mRNA and encodes information for only a single protein and is called monocistronic mRNA. In prokaryotes, clusters of related genes, known as operon, often found next to each other on the chromosome are transcribed together to give a single mRNA and hence are polycistronic.

Before starting transcription, RNA polymerase binds to the promoter, a recognition sequence in front of the gene. Bacterial (prokaryotic) RNA polymerase consists of the core enzyme and the sigma subunit. The core enzyme (2α, β, β^1 and ω) is responsible for RNA synthesis whereas a sigma subunit is responsible for recognition of the promoter. Promoter sequences vary in different organisms. RNA polymerase opens up the DNA to form the transcription bubble. The core enzyme moves ahead, manufacturing RNA leaving the sigma subunit behind at the promoter region. The end of a gene is marked by a terminator sequence that forms a hair pin structure in the RNA. The sub-class of terminators require a recognition protein, known as rho (ρ), to function.

5.9.2 Process of transcription

In prokaryotes, there are three major types of RNAs: mRNA, tRNA, and rRNA. All three RNAs are needed to synthesize a protein in a cell. The mRNA provides the template, tRNA brings amino acids and reads the genetic code, and rRNAs play structural and catalytic role during translation. There is a single DNA-dependent RNA polymerase that catalyses transcription of all types of RNA. It binds to the promoter and initiates transcription (Initiation). The polymerases binding sites are called promoters. It uses nucleoside triphosphate as substrate and energy source and polymerases in a template depended fashion following the rule of complementarity. After the initiation of transcription, the polymerase continues to elongate the RNA, adding one nucleotide after another to the growing RNA chain. Only a short stretch of RNA remains bound to the enzyme, when the polymerase reaches a terminator at the end of a gene, the nascent RNA falls off, so also the RNA polymerase.

The question is, how the RNA polymerases are able to catalyse the three steps initiation,

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